

Total no. of Pages:

Roll no.

6th Semester
MID-SEMESTER EXAMINATION, FEBRUARY-MARCH, 2025

Course Title: Optimization Techniques in AI & ML
Course Code: EAEPE21

Time: 1.5 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	State No Free Lunch Theorem and explain its utility.	1.5	CO1
1b	Classify different optimization problems and explain in brief.	1.5	CO1
2a	Why is it important to strike right balance between exploitation and exploration in a metaheuristic? Explain with an example.	1.5	CO2
2b	How are optimization algorithms Classified according to the principle of a biological and physical-based algorithms. Give details of one algorithm of each type.	1.5	CO1
3a	Explain the role of pheromones in Ant Colony Optimization in detail.	1.5	CO2
3b	What is Metaheuristic and how it is different from orthodox algorithms?	1.5	CO2
4a	Explain how crossover and mutation operators are used in Genetic Algorithm with example.	1.5	CO1
4b	Write down the generic form of mathematical optimization problems.	1.5	CO1
5a	What do you understand by Global and Local Optima? Explain with an example.	1.5	CO1
5b	What are the merits and demerits of Lightning search algorithm-LSA and Backtracking search algorithm-BSA	1.5	CO1